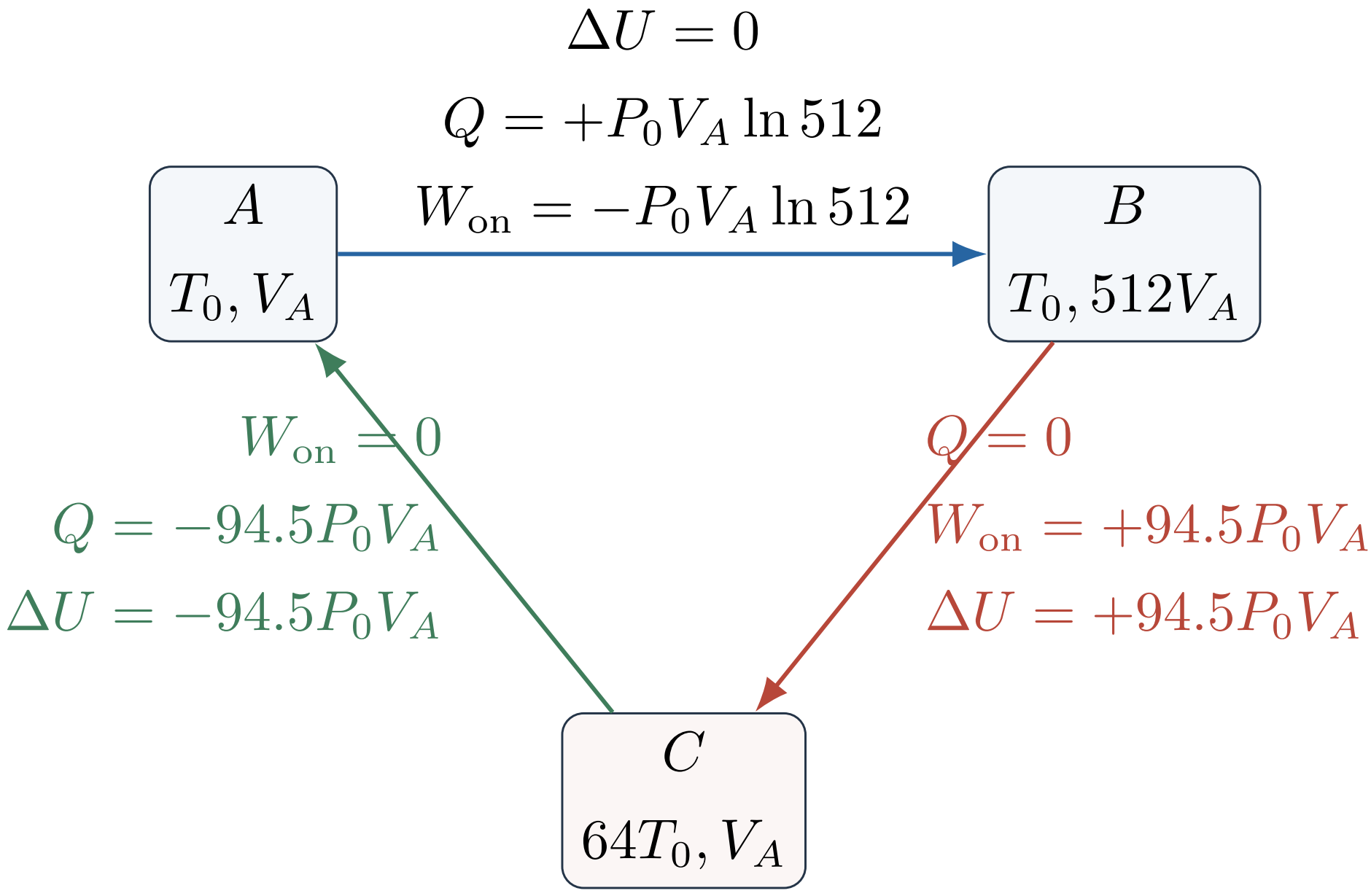

$$\Delta U_{\text{cycle}} = 0, \qquad Q_{\text{cycle}} = -W_{\text{on}}$$